Hardware Business

Our hardware business' revenues are generated from the sales of our Oracle Engineered Systems, server, storage, and industry-specific hardware products. Each hardware product and its related software, such as an operating system or firmware, are highly interdependent and interrelated and are accounted for as a combined performance obligation. The revenues for this combined performance obligation are generally recognized at the point in time that the hardware product and its related software are delivered to the customer and ownership is transferred to the customer. Our hardware business also earns revenues from the sale of hardware support contracts purchased and renewed by our customers at their option and are generally recognized as revenues ratably as the hardware support services are delivered over the contractual term, which is generally one year. The majority of our hardware products are sold through indirect channels such as independent distributors and value-added resellers, and we also market and sell our hardware products through our direct sales force. Operating expenses associated with our hardware business include the cost of hardware products, which consists of expenses for materials and labor used to produce these products by our internal manufacturing operations or by third-party manufacturers, warranty expenses and the impact of periodic changes in inventory valuation, including the impact of inventory determined to be excess and obsolete; the cost of materials used to repair customer products; the cost of labor and infrastructure to provide support services; and sales and marketing expenses, which are largely personnel related and include variable compensation earned by our sales force for the sales of our hardware offerings.

	Year Ended May 31,			
		Percent Change		
(Dollars in millions)	2019	Actual	Constant	2018
Hardware Revenues:				
Americas	$ 1,889	-6%	-4%	$ 2,003
EMEA	1,082	-10%	-5%	1,201
Asia Pacific	733	-7%	-4%	790
Total revenues	3,704	-7%	-5%	3,994
Expenses:				
Hardware products and support (1)	1,327	-14%	-11%	1,547
Sales and marketing (1)	520	-19%	-16%	643
Total expenses (1)	1,847	-16%	-13%	2,190
Total Margin	$ 1,857	3%	6%	$ 1,804
Total Margin %	50%			45%
% Revenues by Geography:				
Americas	51%			50%
EMEA	29%			30%
Asia Pacific	20%			20%

(1) Excludes stock-based compensation and certain expense allocations. Also excludes amortization of intangible assets and certain other GAAP-based expenses, which were not allocated to our operating segment results for purposes of reporting to and review by our CODMs, as further described under "Presentation of Operating Segments and Other Financial Information" above.

Excluding the effects of currency rate fluctuations, total hardware revenues decreased in fiscal 2019 relative to fiscal 2018 due to lower hardware products revenues and, to a lesser extent, lower hardware support revenues. The decrease in hardware products revenues in fiscal 2019 relative to fiscal 2018 was primarily attributable to our continued emphasis on the marketing and sale of our cloud-based infrastructure technologies, which resulted in reduced sales volumes of certain of our hardware product lines and also impacted the volume of hardware support contracts sold in recent periods. This constant currency hardware revenue decrease was partially offset by certain hardware revenue increases related to our Oracle Engineered Systems offerings, primarily Oracle Exadata.

Excluding the effects of currency rate fluctuations, total hardware expenses decreased in fiscal 2019 compared to fiscal 2018 primarily due to lower hardware products and support costs and lower sales and marketing employee related expenses, all of which aligned to lower hardware revenues.

In constant currency, total margin and total margin as a percentage of revenues for our hardware segment increased in fiscal 2019 due to lower expenses .

Services Business

We offer services to customers and partners to help to maximize the performance of their investments in Oracle applications and infrastructure technologies. Services revenues are generally recognized as the services are performed. The cost of providing our services consists primarily of personnel related expenses, technology infrastructure expenditures, facilities expenses and external contractor expenses.

	Year Ended May 31,			
		Percent Change		
(Dollars in millions)	2019	Actual	Constant	2018
Services Revenues:				
Americas	$ 1,576	-5%	-3%	$ 1,654
EMEA	1,021	-2%	2%	1,046
Asia Pacific	643	-7%	-4%	695
Total revenues	3,240	-5%	-2%	3,395
Total Expenses (1)	2,703	-1%	2%	2,729
Total Margin	$ 537	-19%	-18%	$ 666
Total Margin %	17%			20%
% Revenues by Geography:				
Americas	49%			49%
EMEA	31%			31%
Asia Pacific	20%			20%

(1) Excludes stock-based compensation and certain expense allocations. Also excludes amortization of intangible assets and certain other GAAP-based expenses, which were not allocated to our operating segment results for purposes of reporting to and review by our CODMs, as further described under "Presentation of Operating Segments and Other Financial Information" above.

Excluding the effects of currency rate fluctuations, our total services revenues decreased in fiscal 2019 relative to fiscal 2018 primarily due to revenue declines in our education services and, to a lesser extent, our consulting services. During fiscal 2019, constant currency increases in our EMEA-based services revenues were offset by constant currency services revenue decreases in the Americas and the Asia Pacific regions.

In constant currency, total services expenses increased in fiscal 2019 compared to fiscal 2018 primarily due to an increase in employee related expenses and external contractor expenses associated with investments in our consulting services that support our cloud offerings. In constant currency, total margin and total margin as a percentage of total services revenues decreased during fiscal 2019 relative to fiscal 2018 due to decreased revenues and increased expenses for this business.

Research and Development Expenses: Research and development expenses consist primarily of personnel related expenditures. We intend to continue to invest significantly in our research and development efforts because, in our judgment, they are essential to maintaining our competitive position.

	Year Ended May 31,			
		Percent Change		
(Dollars in millions)	2019	Actual	Constant	2018
Research and development (1)	$ 5,063	-2%	0%	$ 5,163
Stock-based compensation	963	5%	5%	921
Total expenses	$ 6,026	-1%	0%	$ 6,084
% of Total Revenues	15%			15%

(1) Excluding stock-based compensation